

# Seonghwan Seo

Ph.D. Candidate | Korea Advanced Institute of Science and Technology

[shwan0106@kaist.ac.kr](mailto:shwan0106@kaist.ac.kr) | [Website](#) | [Google Scholar](#) | [GitHub](#)

## RESEARCH INTERESTS

---

### Deep Learning

Generative Flow Network (GFlowNet); Graph Neural Network; 3D voxel image modeling

### Small Molecule Drug Discovery

Generative Model; Synthesizability; Virtual screening

### Protein Discovery

Structure prediction (co-folding); Protein design

## EXPERIENCES

---

### Intelligent Chemistry Lab, KAIST

AI for Drug Discovery

Supervisor: Woo Youn Kim

Daejeon, South Korea

June 2019 - Present

- Development of an unbiased drug-likeness scoring
- Development of a DL framework for fully-automated receptor-based pharmacophore modeling
- Research projects on synthesizable drug discovery
- Development of a trillion-scale library search technology with eMolecules & HITS
- National project for next-generation bio foundation model K-Fold

### HITS Inc.

AI Research Intern

Manager: Jaechang Lim

Seoul, South Korea

Dec. 2020 - Aug. 2022

- Development of reaction template-based generative model for synthesizable molecular design with desired property
- Development of transformer-based model for synthesis path generation

## EDUCATION

---

### Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. student in Chemistry

Supervisor: Woo Youn Kim

Daejeon, South Korea

Aug. 2022 - Feb. 2028 (expected)

### Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Chemistry & Computer Science (Double Major)

Supervisor: Young Min Rhee

GPA: 3.99/4.3

Daejeon, South Korea

Feb. 2018 - Aug. 2022

## PUBLICATIONS

---

\* indicates equal contribution

- Compositional Flows for 3D Molecule and Synthesis Pathway Co-design [[link](#)]  
Tony Shen\*, **Seonghwan Seo**\*, Ross Irwin, Kieran Didi, Simon Olsson, Woo Youn Kim, Martin Ester  
*International Conference on Machine Learning (ICML), 2025*
- Generative Flows on Synthetic Pathway for Drug Design [[link](#)]  
**Seonghwan Seo**, Minsu Kim, Tony Shen, Martin Ester, Jinkyoo Park, Sungsoo Ahn, Woo Youn Kim  
*International Conference on Learning Representations (ICLR), 2025*
- PharmacoNet: deep learning-guided pharmacophore modeling for ultra-large-scale virtual screening [[link](#)]  
**Seonghwan Seo**, Woo Youn Kim  
*Chemical Science, 2024*
- TacoGFN: Target Conditioned GFlowNet for Structure-based Drug Discovery [[link](#)]  
Tony Shen, **Seonghwan Seo**, Grayson Lee, Mohit Pandey, Jason R Smith, Artem Cherkasov, Woo Youn Kim, Martin Ester  
*Transactions on Machine Learning Research (TMLR), 2024*
- Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly [[link](#)] [[cover](#)]  
**Seonghwan Seo**, Jaechang Lim, Woo Youn Kim  
*Advanced Science, 2023 (selected for a journal cover)*
- Drug-likeness scoring based on unsupervised learning [[link](#)]

## PRESENTATIONS

---

### Invited

- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"  
Oral workshop and tutorial at *AI-BIO, Artificial Intelligence Institute, Seoul National University*, Seoul, South Korea (Nov. 2022)

### Contributed

- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"  
Oral spotlight presentation at *ICLR 2025 Workshop on AI for Accelerated Materials Discovery (AI4Mat)*, Singapore. (Apr. 2025)
- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"  
Oral spotlight presentation at *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"  
Poster presentation at *ICLR 2025*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"  
Poster presentation at *NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)*, Vancouver, Canada. (Dec. 2024)
- "PharmacoNet: Accelerating Large-Scale Virtual Screening by Deep Pharmacophore Modeling"  
Poster presentation at *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development (AI4D3)*, New Orleans, LA, USA. (Dec. 2023)
- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"  
Poster presentation at *2023 Accelerate Conference*, Toronto, Canada (Aug. 2023)

## AWARDS & SCHOLARSHIPS

---

- **President of TTA Award**  
*Korea AI-Champion Competition, 2025*
- **KAIST Alumni Scholarship**  
*KAIST Alumni Association, 2021 - 2022*
- **Younghoon Lee Scholarship**  
*Department of Chemistry, KAIST, 2021*
- **Model Student Awards**  
*KAIST, 2021*
- **Dean's List**  
*College of Natural Science, KAIST, 2021*
- **National Science & Technology Scholarship**  
*Government of South Korea, 2020 - 2021*
- **Dean's List**  
*School in Freshman, KAIST, 2019*

## ACADEMIC REVIEWER

---

- International Conference on Learning Representation (ICLR), 2026
- International Conference on Learning Representation (ICLR), 2025
- Nature Communication, 2024

## SKILLS

---

### Languages

Korean (native), English (upper-intermediate)

### Deep Learning Tools

Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Numba, Pandas

### Chemistry Tools

RDKit, OpenBabel, Gemmi, Vina, Smina, rDock